Capability (d_R, d_B) = (0, 1) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (0, 2) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (0, 3) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (0, 4) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (0, 5) · Unseen: both delays novel

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (0, 6) · Unseen: both delays novel

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (0, 7) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (0, 8) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (0, 9) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (1, 0) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (1, 2) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (1, 3) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (1, 4) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Down / — · Right / — · Left / — · Stay / —

Up / R · Down / R · Right / R · Left / R · Stay / R

Up / B · Down / B · Right / B · Left / B · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (1, 5) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (1, 6) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (1, 7) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — | Down / — | Right / — | Left / — | Stay / —
Up / R | Down / R | Right / R | Left / R | Stay / R
Up / B | Down / B | Right / B | Left / B | Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (1, 8) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1 matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

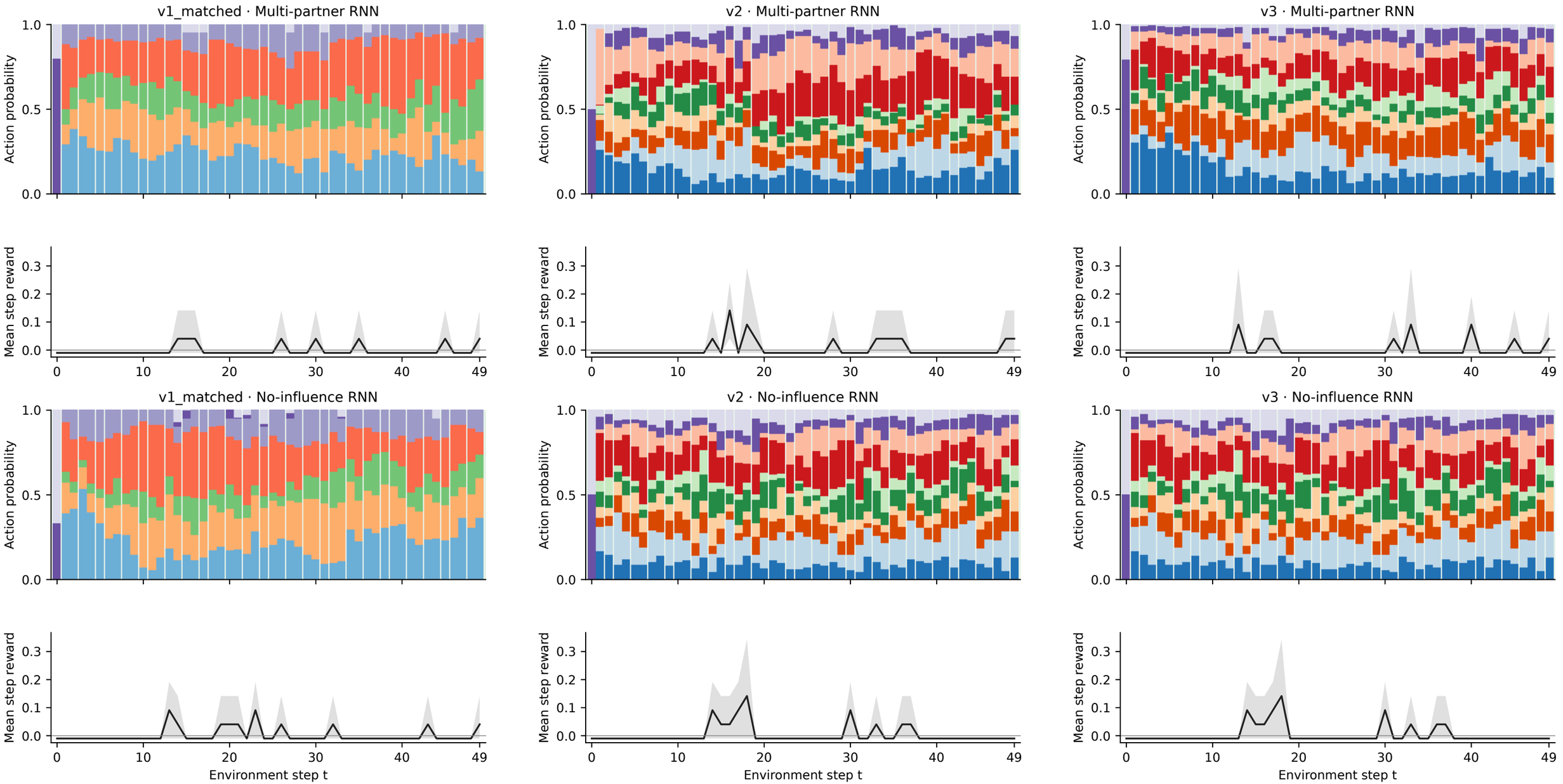
Up / — · Down / — · Right / — · Left / — · Stay / —

Up / R · Down / R · Right / R · Left / R · Stay / R

Up / B · Down / B · Right / B · Left / B · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (1, 9) · Seen: diverse training



Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (2, 0) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (2, 1) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (2, 3) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

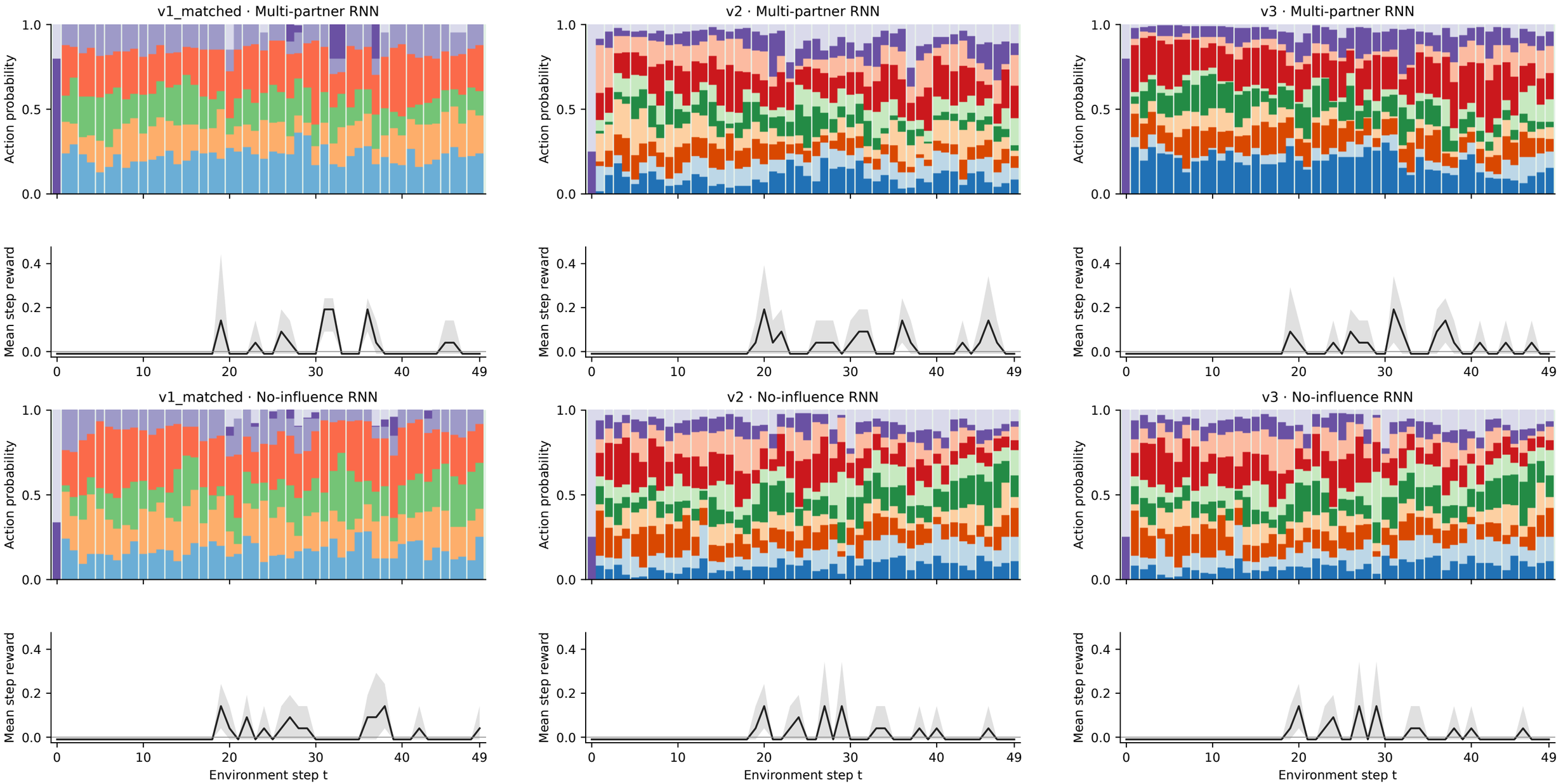
Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (2, 4) · Seen: diverse training



Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (2, 5) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (2, 6) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (2, 7) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (2, 8) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

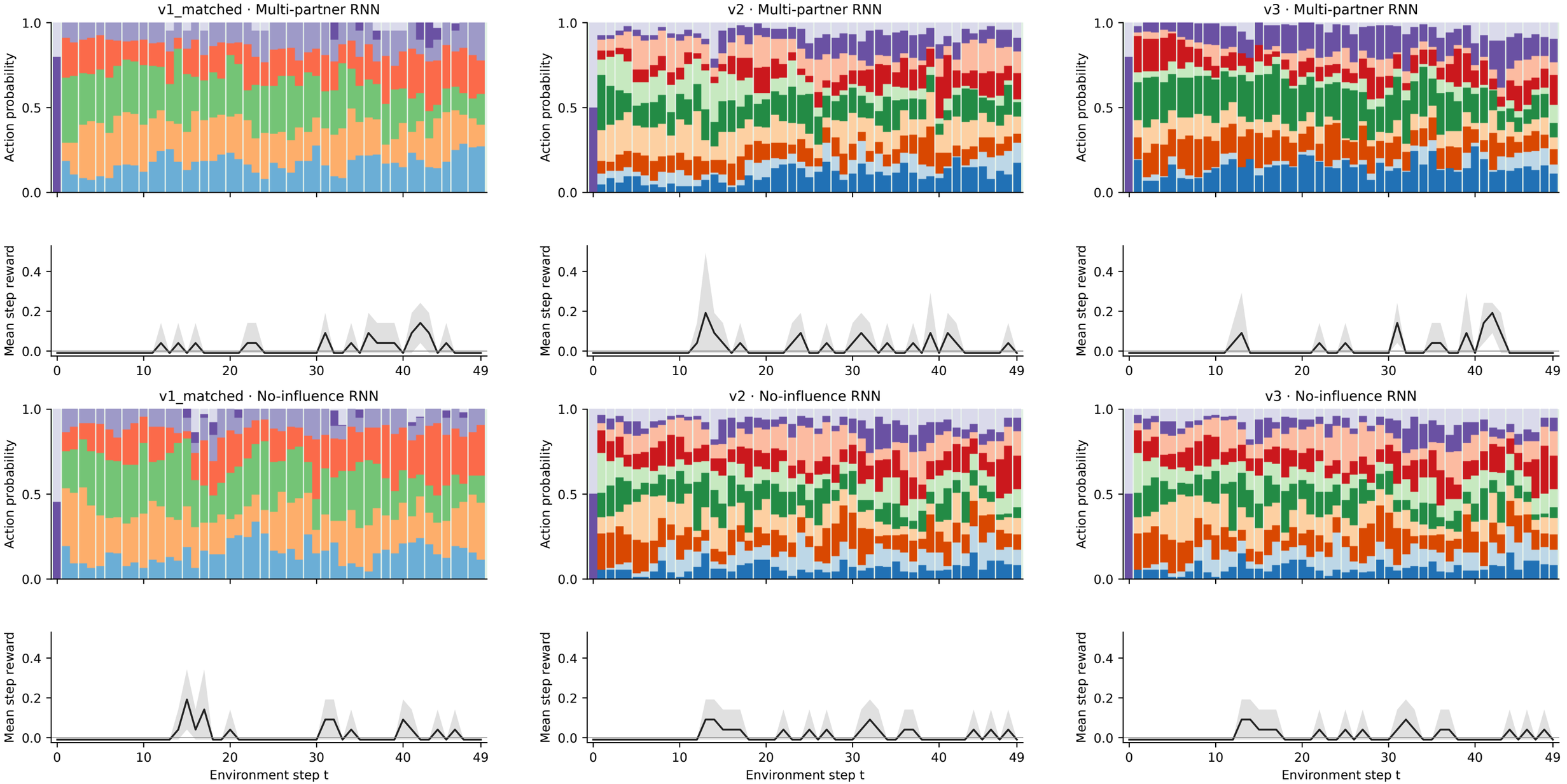
Up / — · Down / — · Right / — · Left / — · Stay / —

Up / R · Down / R · Right / R · Left / R · Stay / R

Up / B · Down / B · Right / B · Left / B · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (2, 9) · Seen: diverse training



Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (3, 0) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

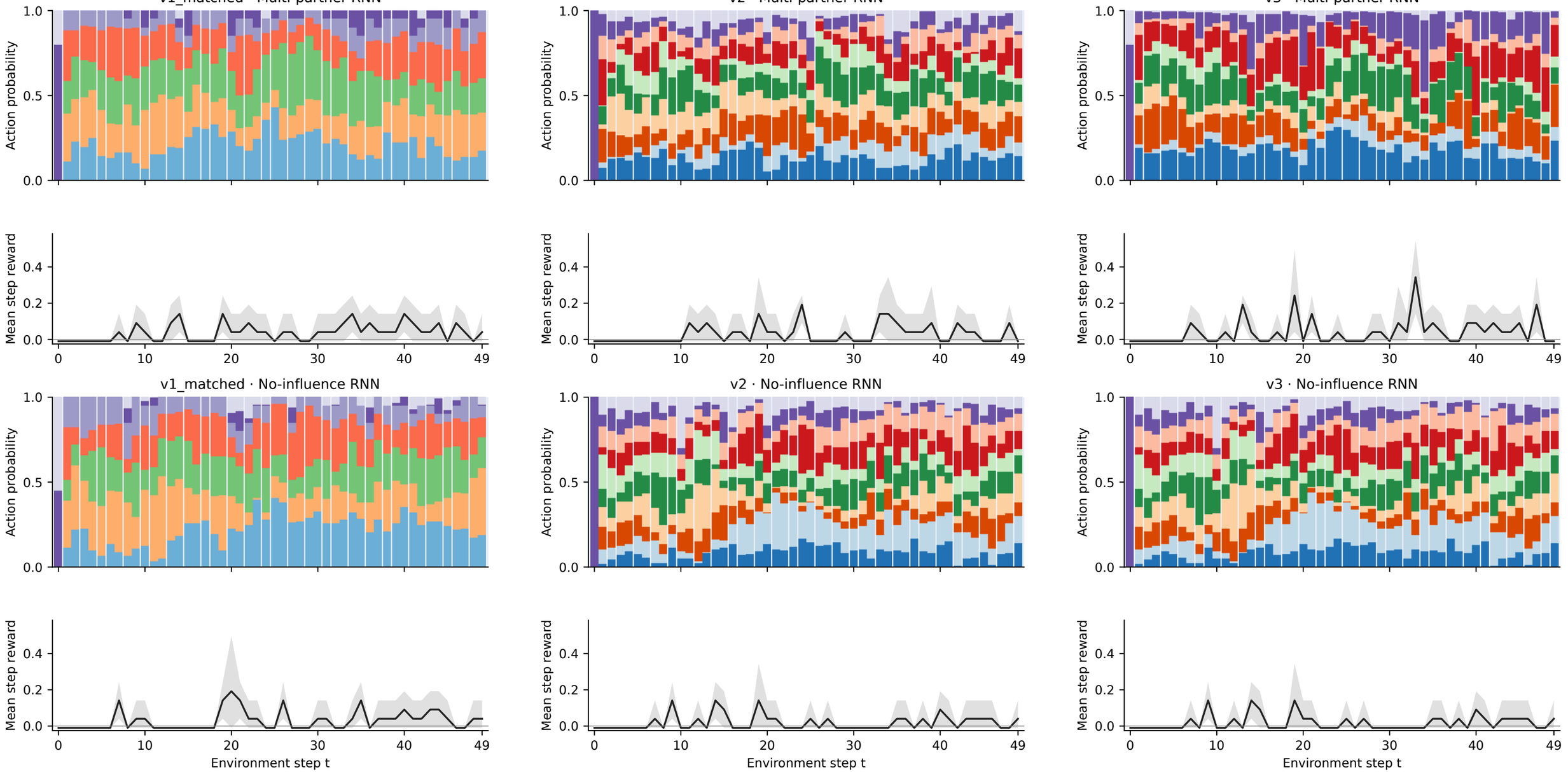
Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (3, 1) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN



Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (3, 2) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

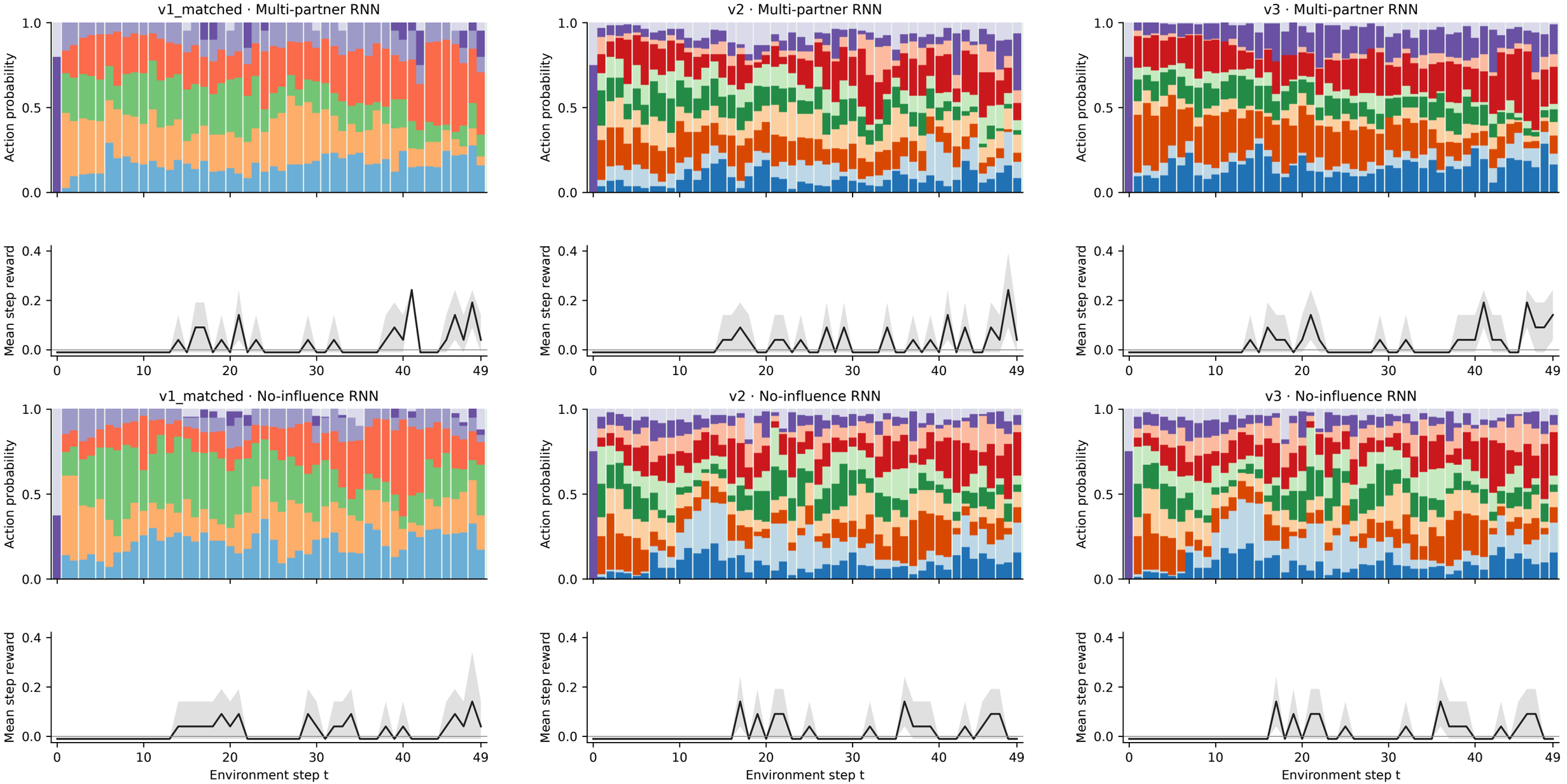
Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (3, 4) · Seen: diverse training



Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (3, 5) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (3, 6) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (3, 7) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Down / — · Right / — · Left / — · Stay / —

Up / R · Down / R · Right / R · Left / R · Stay / R

Up / B · Down / B · Right / B · Left / B · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (3, 8) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — | Down / — | Right / — | Left / — | Stay / —

Up / R | Down / R | Right / R | Left / R | Stay / R

Up / B | Down / B | Right / B | Left / B | Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (3, 9) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — | Down / — | Right / — | Left / — | Stay / —

Up / R | Down / R | Right / R | Left / R | Stay / R

Up / B | Down / B | Right / B | Left / B | Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (4, 0) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (4, 1) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (4, 2) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (4, 3) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — | Down / — | Right / — | Left / — | Stay / —

Up / R | Down / R | Right / R | Left / R | Stay / R

Up / B | Down / B | Right / B | Left / B | Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (4, 5) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

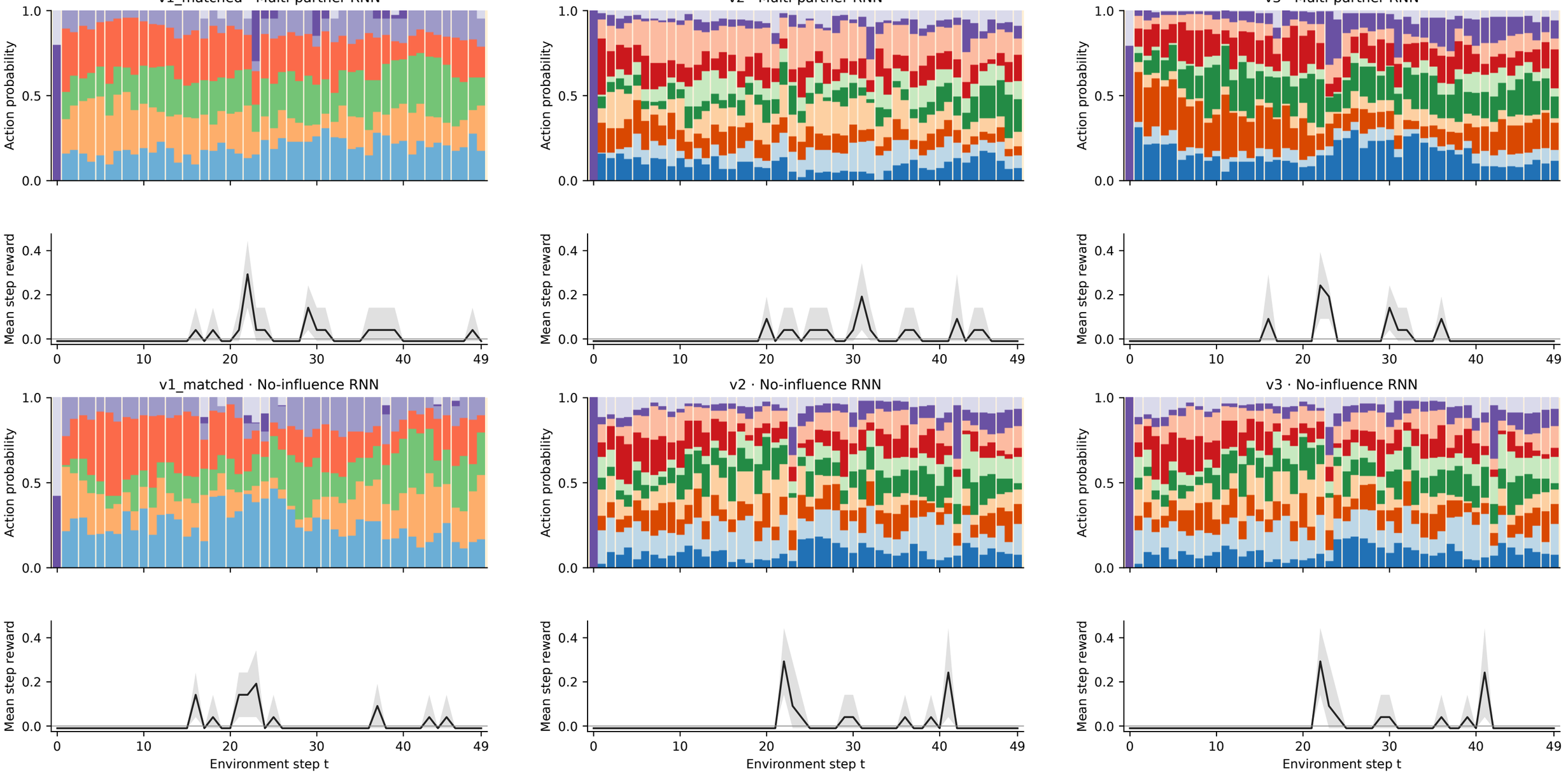
Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (4, 6) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN



Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (4, 7) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (4, 8) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (4, 9) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Down / — · Right / — · Left / — · Stay / —

Up / R · Down / R · Right / R · Left / R · Stay / R

Up / B · Down / B · Right / B · Left / B · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (5, 0) · Unseen: both delays novel

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (5, 1) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1 matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — | Down / — | Right / — | Left / — | Stay / —
Up / R | Down / R | Right / R | Left / R | Stay / R
Up / B | Down / B | Right / B | Left / B | Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (5, 2) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

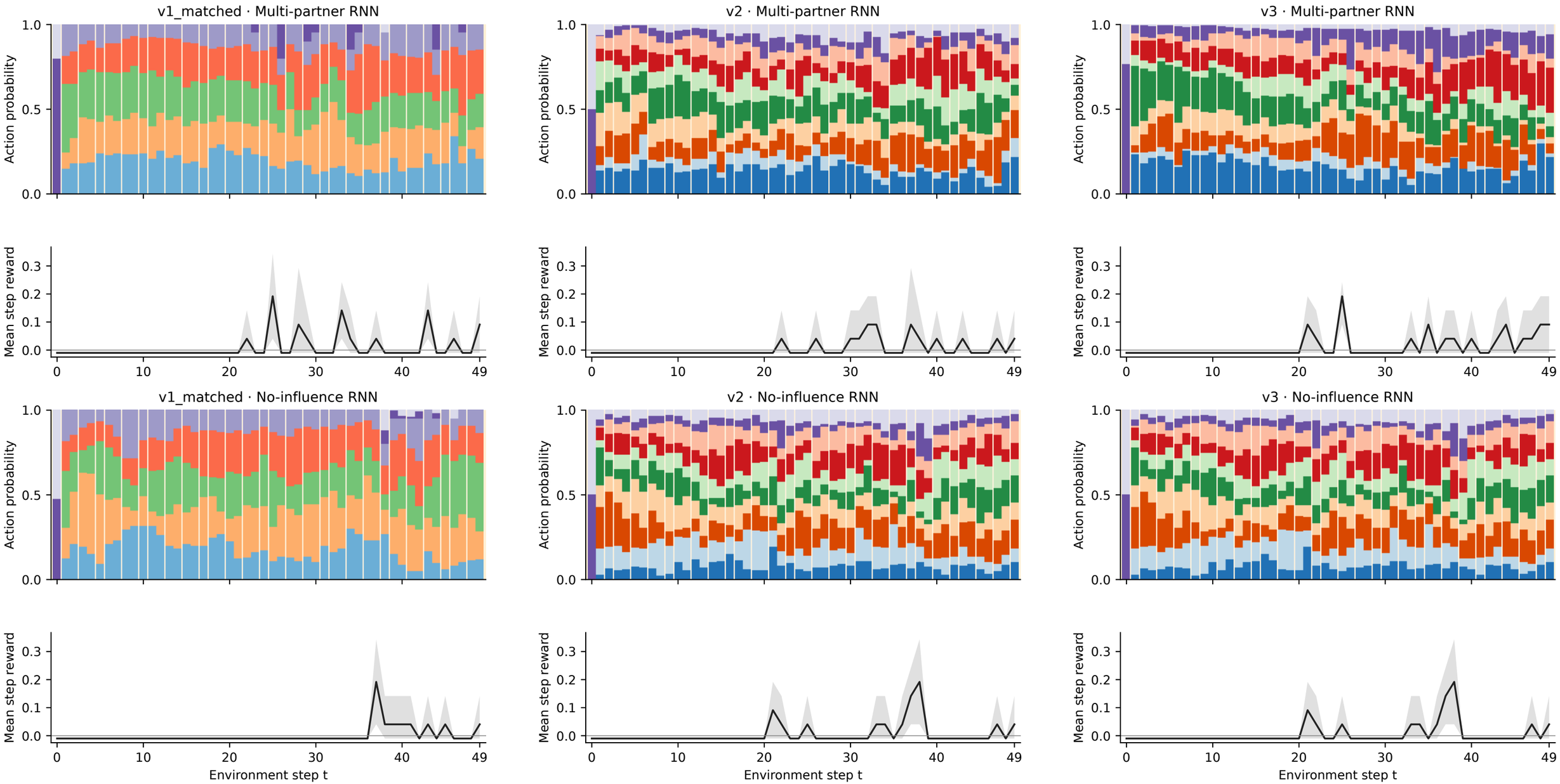
Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (5, 3) · Unseen: one novel delay



Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (5, 4) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — | Down / — | Right / — | Left / — | Stay / —
Up / R | Down / R | Right / R | Left / R | Stay / R
Up / B | Down / B | Right / B | Left / B | Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (5, 6) · Unseen: both delays novel

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — | Down / — | Right / — | Left / — | Stay / —
Up / R | Down / R | Right / R | Left / R | Stay / R
Up / B | Down / B | Right / B | Left / B | Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (5, 7) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (5, 8) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1 matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (5, 9) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — | Down / — | Right / — | Left / — | Stay / —

Up / R | Down / R | Right / R | Left / R | Stay / R

Up / B | Down / B | Right / B | Left / B | Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (6, 0) · Unseen: both delays novel

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

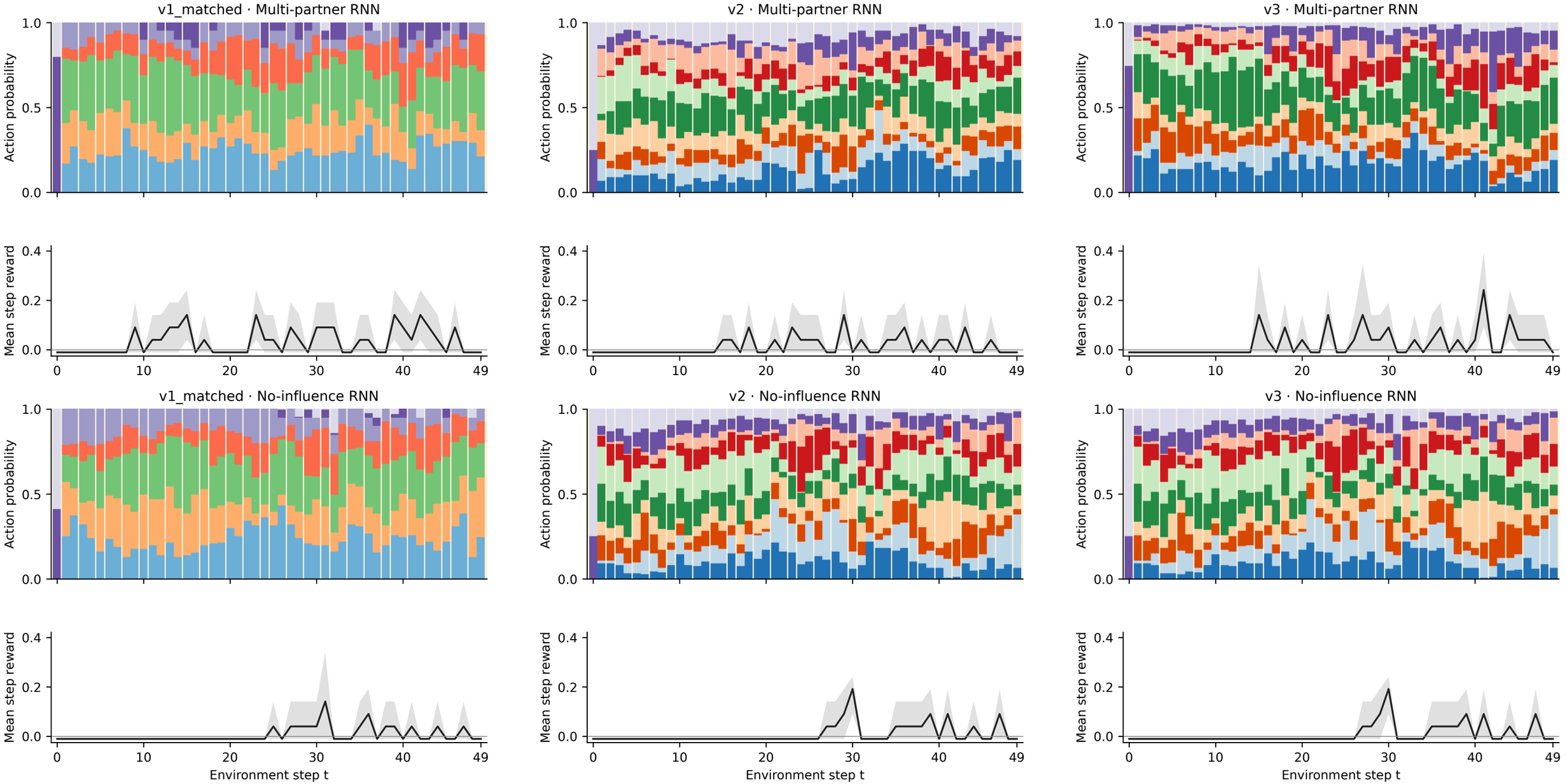
Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (6, 1) · Unseen: one novel delay



Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (6, 2) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (6, 3) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (6, 4) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (6, 5) · Unseen: both delays novel

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (6, 7) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (6, 8) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

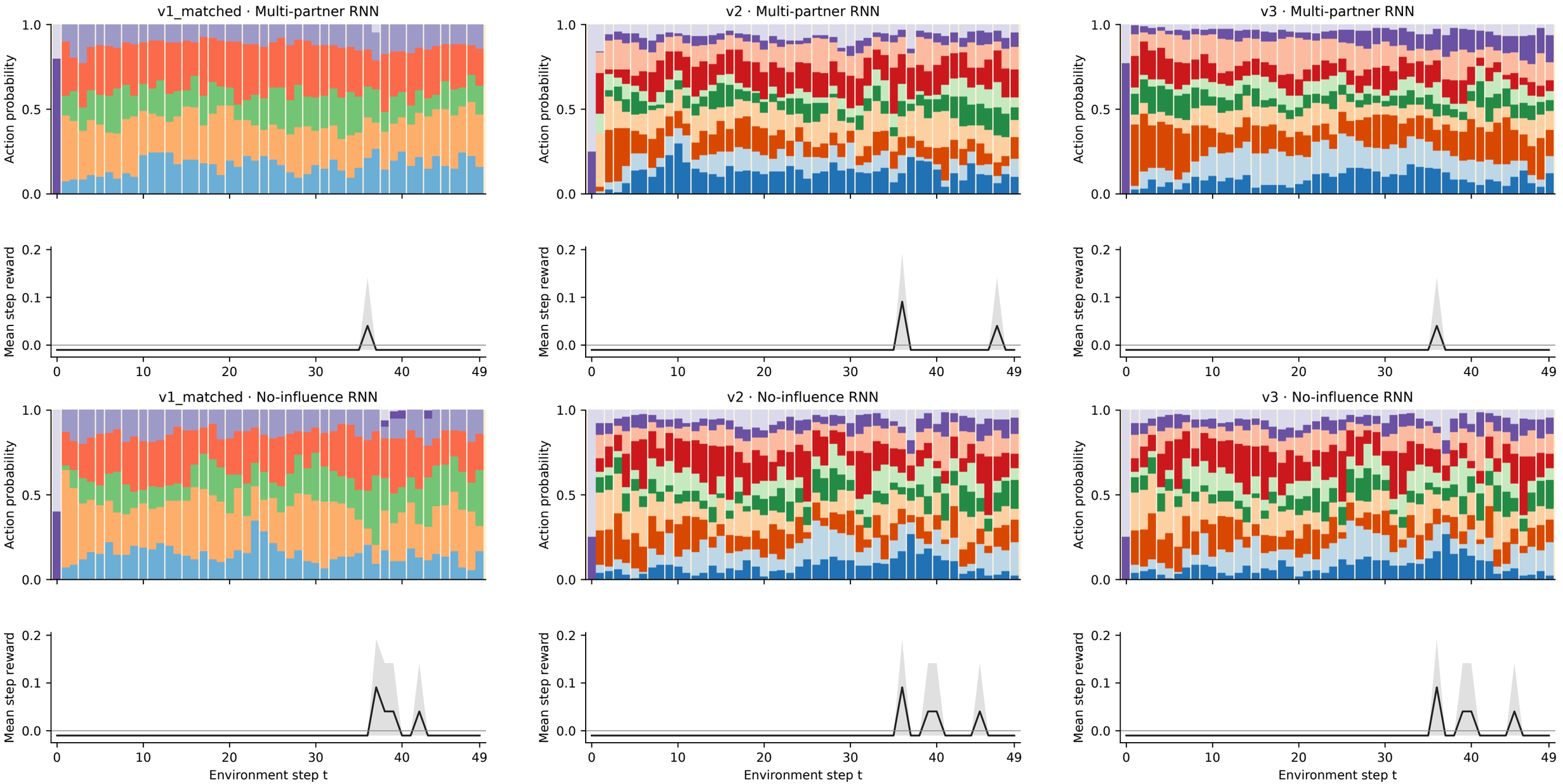
Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (6, 9) · Unseen: one novel delay



Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (7, 0) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (7, 1) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1 matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

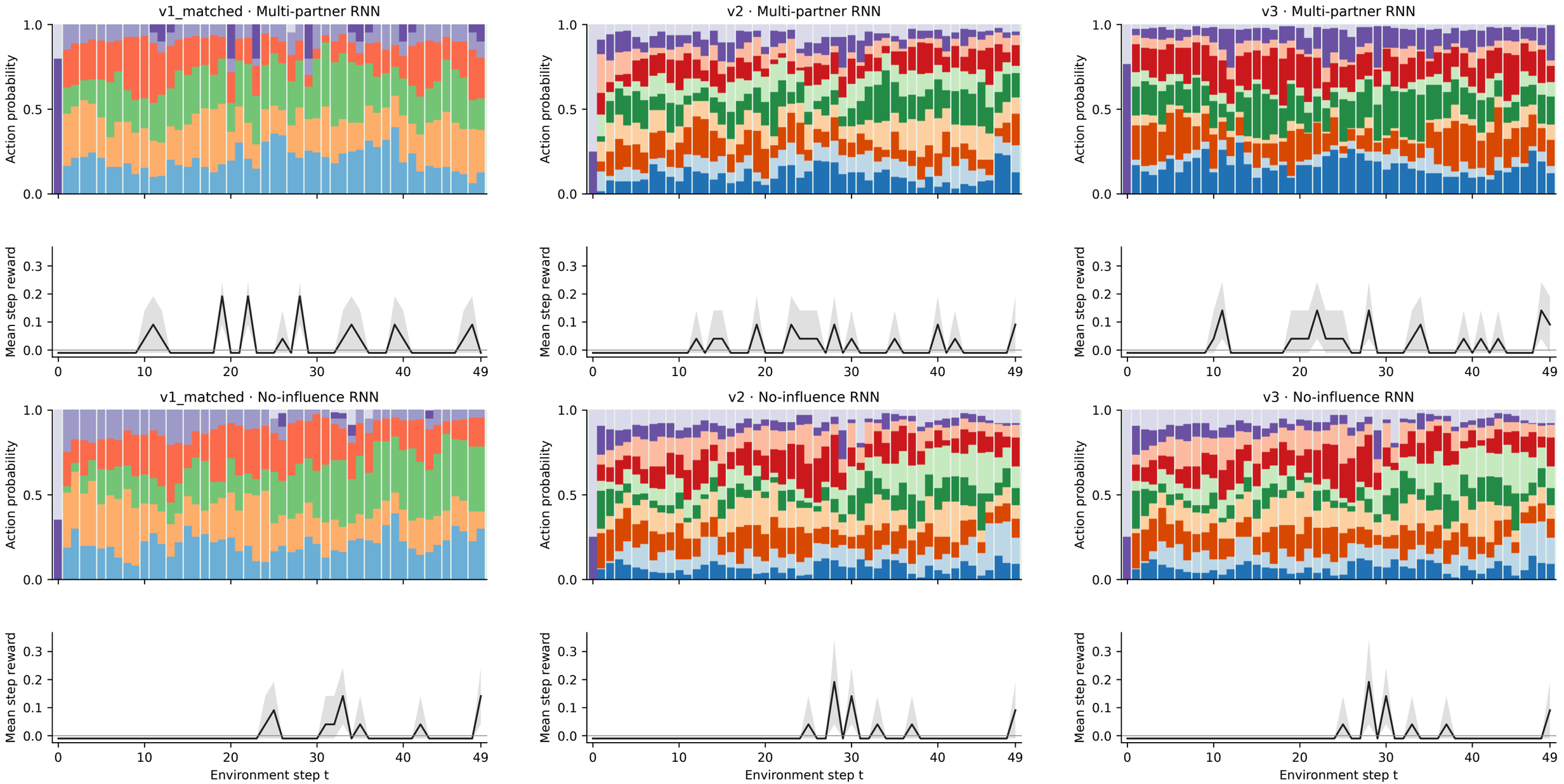
Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (7, 2) · Seen: diverse training



Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (7, 3) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1 matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (7, 4) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (7, 5) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (7, 6) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (7, 8) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (7, 9) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (8, 0) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (8, 1) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (8, 2) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1 matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — | Down / — | Right / — | Left / — | Stay / —

Up / R | Down / R | Right / R | Left / R | Stay / R

Up / B | Down / B | Right / B | Left / B | Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (8, 3) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Down / — · Right / — · Left / — · Stay / —

Up / R · Down / R · Right / R · Left / R · Stay / R

Up / B · Down / B · Right / B · Left / B · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (8, 4) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

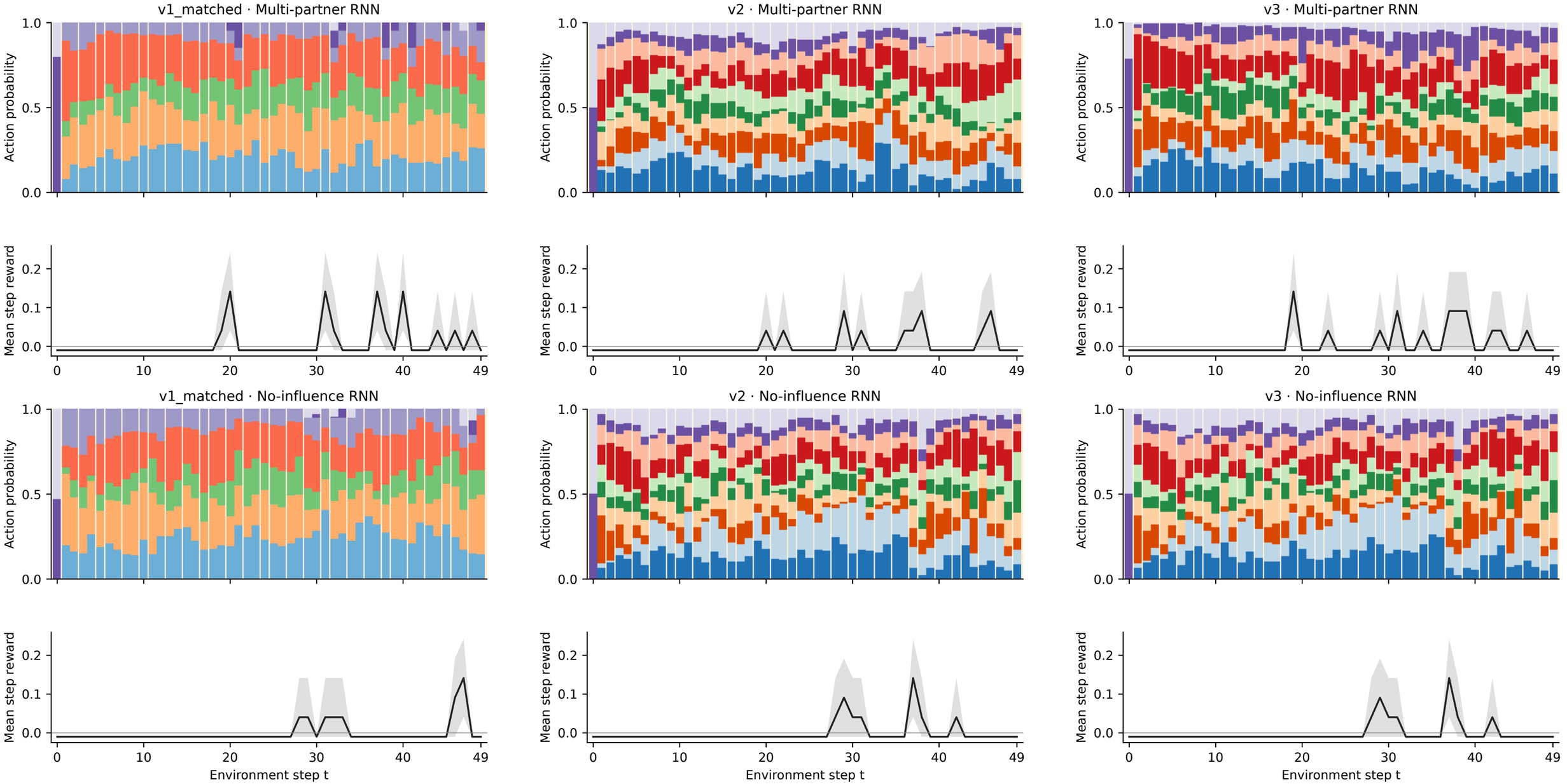
Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (8, 5) · Unseen: one novel delay



Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (8, 6) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (8, 7) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (8, 9) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (9, 0) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (9, 1) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (9, 2) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (9, 3) · Seen: diverse training

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Down / — · Right / — · Left / — · Stay / —

Up / R · Down / R · Right / R · Left / R · Stay / R

Up / B · Down / B · Right / B · Left / B · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (9, 4) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

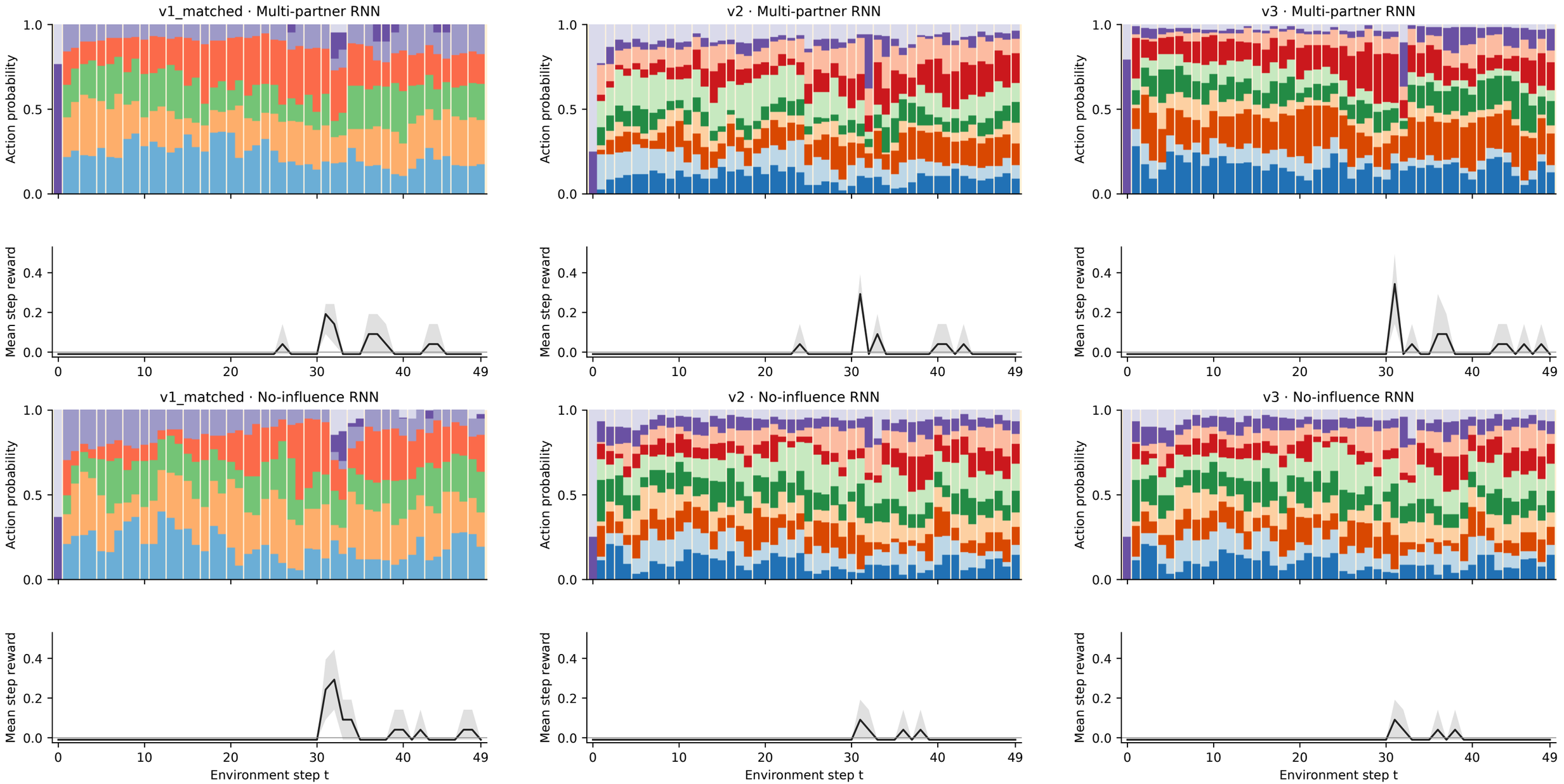
Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (9, 5) · Unseen: one novel delay



Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (9, 6) · Unseen: one novel delay

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (9, 7) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / — · Up / R · Up / B · Down / — · Down / R · Down / B · Right / — · Right / R · Right / B · Left / — · Left / R · Left / B · Stay / — · Stay / R · Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.

Capability (d_R, d_B) = (9, 8) · Unseen pair; familiar delays

v1_matched · Multi-partner RNN

v2 · Multi-partner RNN

v3 · Multi-partner RNN

Action probability

Mean step reward

v1_matched · No-influence RNN

v2 · No-influence RNN

v3 · No-influence RNN

Environment step t

Joint action = movement / requested ego allocation (—: none; R: red; B: blue)

Up / —	Down / —	Right / —	Left / —	Stay / —
Up / R	Down / R	Right / R	Left / R	Stay / R
Up / B	Down / B	Right / B	Left / B	Stay / B

Probabilities reconstructed from saved h_t + frozen actor; rewards from saved transitions. 4 test episodes/profile/seed × 5 seeds. Reward shading: 95% seed bootstrap CI. No time smoothing.